Day 4 Cheat Sheet: A/B Testing, Sample Size & Power

Design first, run to a fixed n, analyze once

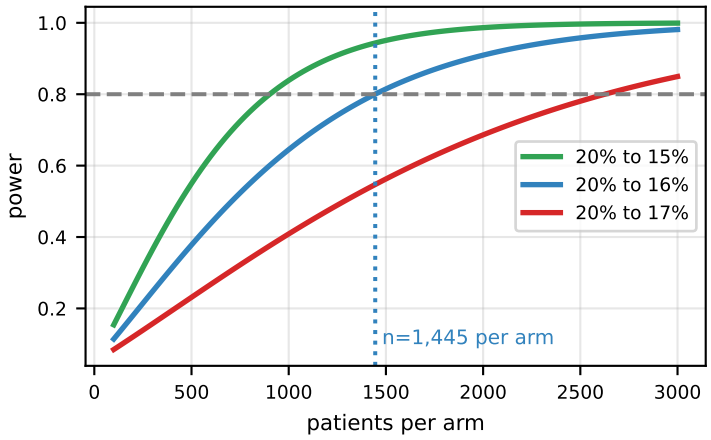
A/B TEST = randomized experiment: control (A) vs. change (B), one primary metric
- Decide BEFORE launch: metric, minimum effect worth detecting (MDE), alpha, power, n
- Randomize patients (or clinics) so the only systematic difference is the change

FOUR OUTCOMES	reality: no effect	reality: real effect
test says "effect"	Type I error (alpha, 5%)	correct (power, 80%)
test says "no effect"	correct (95%)	Type II error (beta, 20%)

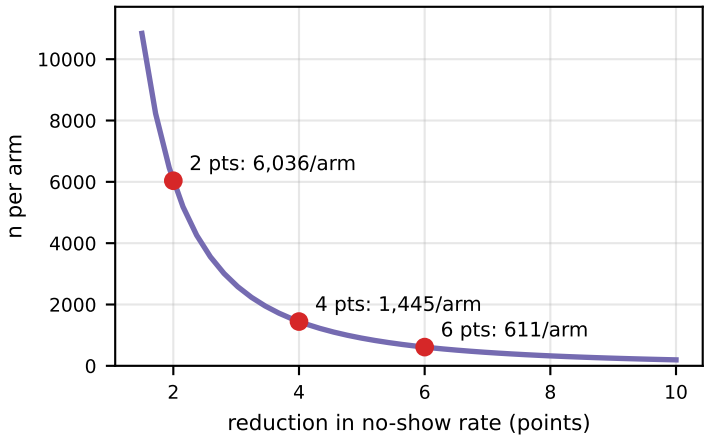
SAMPLE SIZE PER ARM (two rates): $n = (z_a \sqrt{2\,p(1-p)} + z_b \sqrt{p_1q_1 + p_2q_2})^2 / (p_1-p_2)^2$
- $z_a = 1.96$ (alpha 0.05, two-sided), $z_b = 0.84$ (80% power), p = average of p_1 and p_2
- Halving the effect you want to detect needs about 4x the patients

PEEKING: checking p after every batch and stopping at $p < 0.05$ breaks the 5% promise.
Fix: pick n in advance and look once, or use a planned sequential design.

Power vs. patients per arm (no-show reduction)



Smaller effects need far more patients (80% power)



THIS DATA (simulated): clinic no-show rate 20%, SMS reminder hoped to cut it to 16%
n per arm for 80% power: formula 1,447, statsmodels 1,445 | run with 1,450 per arm
Result: SMS 15.7% vs control 20.1%, difference -4.3 pts, 95% CI [-7.1, -1.6], p = 0.0023
Simulated power at 1,450: 79.5% | at 500 per arm: 37.0% | 15 peeks, no real effect: 23.7% false positives

COMMON MISTAKES
- No power calculation: a 'no significant difference' from a tiny test means very little
- Stopping as soon as $p < 0.05$, or extending the test until it is significant
- Testing 10 metrics and reporting the one that 'won' (pick ONE primary metric up front)
- Reporting p alone: give the difference with its 95% CI (Day 3)

```
from statsmodels.stats.power import NormalIndPower
from statsmodels.stats.proportion import proportion_effectsize, proportions_ztest
h = proportion_effectsize(0.20, 0.16)                  # Cohen's h
NormalIndPower().solve_power(h, alpha=0.05, power=0.8)  # n per arm
z, p = proportions_ztest([x_trt, x_ctrl], [n_trt, n_ctrl])
confint_proportions_2indep(x_trt, n_trt, x_ctrl, n_ctrl)  # CI for the gap
```

Infographic: The Life of a Good A/B Test

Simulated clinic test: SMS appointment reminders vs. standard reminders

Hypothesis — SMS cuts no-shows

> **Power calc** — MDE 4 pts, 1,450/arm

> **Randomize** — patients to A or B

> **Run** — to the fixed sample size

> **Analyze once** — z-test + 95% CI

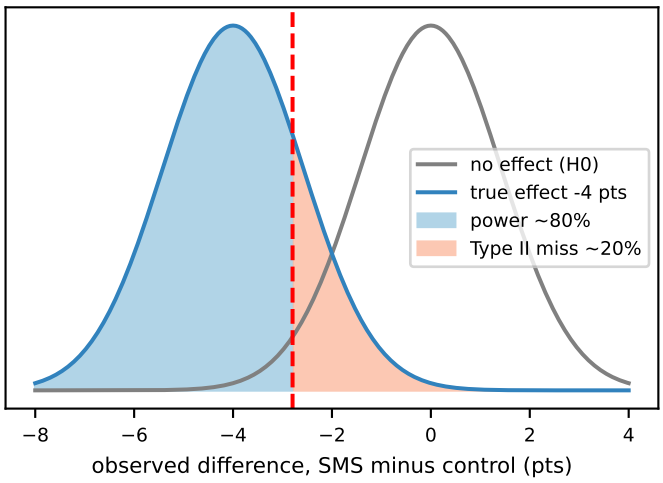
ALPHA
5%
false alarm

POWER
80%
catch it

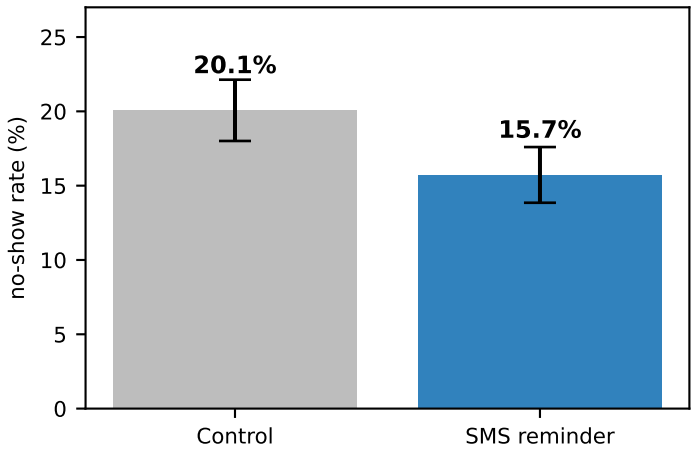
N PER ARM
1,450
patients

PEEKING x15
24%
false alarm

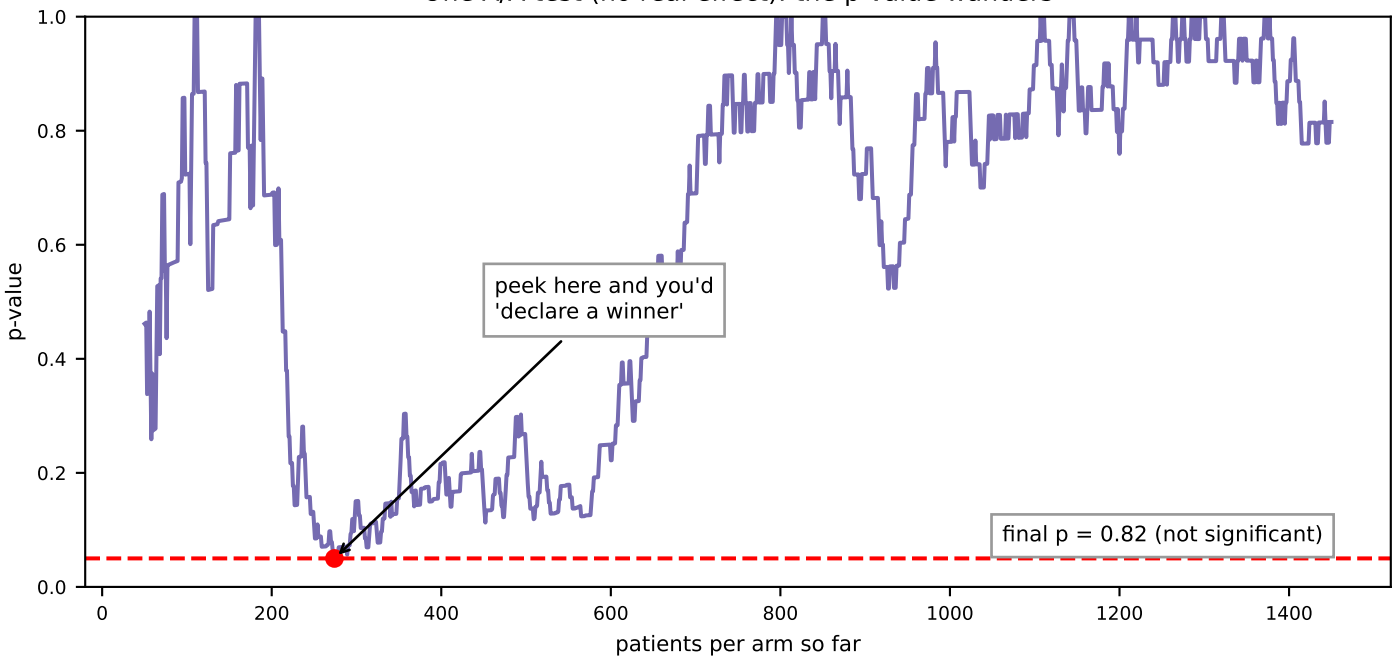
Power = chance the observed gap clears the cutoff



Result: -4.3 pts, 95% CI [-7.1, -1.6], p = 0.002



One A/A test (no real effect): the p-value wanders



Decide n before you start and look once. A p-value you watch every day will dip under 0.05 by chance.

Hands-On Worksheet: ds-genai-90-days / day04

60-90 minutes | notebook: day04/ab_test_power.ipynb

[ ] mkdir day04, open VS Code, conda activate ds90, conda install statsmodels

[ ] Write the test plan: metric, MDE (20% to 16% no-shows), alpha 0.05, power 0.80

[ ] Compute n per arm by formula and with NormalIndPower (expect about 1,445-1,447)

[ ] Simulate the test with 1,450 per arm (seed 4) and run proportions_ztest

[ ] Report the difference with a 95% CI (confint_proportions_2indep)

[ ] Simulate 2,000 trials: power at 1,450 vs. 500 per arm, and an A/A test

[ ] Peeking: 15 looks at an A/A test, stop at p < 0.05. Count false positives

[ ] Plot power curves for 3, 4 and 5 point reductions

[ ] Write a 1-paragraph decision memo: recommend SMS or not, with the CI

[ ] git add day04 && git commit -m 'Day 4: A/B testing and power' && git push

[ ] Stretch: Snowflake: no-show rate and z-statistic by arm from appointments

```
import numpy as np
from statsmodels.stats.power import NormalIndPower
from statsmodels.stats.proportion import proportion_effectsize, proportions_ztest
h = proportion_effectsize(0.20, 0.16)
print(NormalIndPower().solve_power(effect_size=h, alpha=0.05, power=0.80))  # ~1445

rng = np.random.default_rng(4); n = 1450
ctrl = rng.random(n) < 0.20; trt = rng.random(n) < 0.16
z, p = proportions_ztest([trt.sum(), ctrl.sum()], [n, n])
print(trt.mean(), ctrl.mean(), p)        # expect 0.157, 0.201, 0.0023
```

n per arm: _____ Difference: _____ pts 95% CI: [_____, _____]

Power at 500/arm: _____ False positive rate with 15 peeks: _____

Day 4 Self-Check Quiz

1. In plain words, what is statistical power?

_____

2. Which error is a false alarm: Type I or Type II? Which one does alpha control?

_____

3. You want to detect a 2-point drop instead of 4. Roughly how many more patients?

_____

4. A 500-per-arm test finds 'no significant difference'. Why is that weak evidence?

_____

5. Why randomize patients instead of giving SMS to the clinic that asks for it?

_____

6. A manager checks the dashboard daily and stops at $p < 0.05$. What goes wrong?

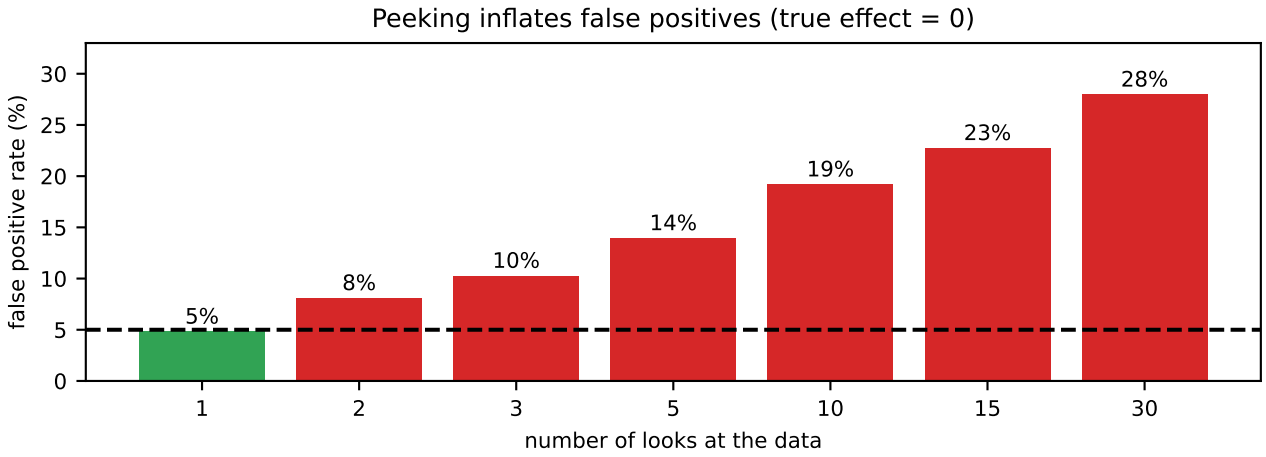
_____

7. The result is -4.3 pts, 95% CI [-7.1, -1.6]. Write one sentence for the COO.

_____

8. Name two things you must decide before the test starts.

_____



Answers: 1 the chance the test detects a real effect of the size you care about 2 Type I; alpha 3 about 4x (halving the effect quadruples n)
4 power was only ~37%, so a real 4-pt effect would be missed most of the time 5 so the groups differ only by the change (no confounding)
6 false positive rate rises far above 5% (about 24% with 15 looks) 7 SMS cut no-shows by about 4 pts, plausibly 1.6 to 7.1 (simulated data)
8 primary metric, MDE, alpha, power, sample size, analysis plan